PARENT PORTAL MODULE

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · School Administrators · Parents · IT Team

Related Modules:
Students · Academics · Attendance · Finance · Communication · Health · Transport · Cafeteria

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1. MODULE OVERVIEW

1.1 Purpose

The **Parent Portal Module** provides parents/guardians with **real-time visibility and interaction** with their child's school life.

It enables:

- Monitoring academic performance
- Tracking attendance and behavior
- Managing school fees and payments
- Receiving updates and communicating with staff
- Accessing services (transport, health, cafeteria, events)

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2. CORE OBJECTIVES

1. Provide **centralized access** to student data
2. Enable **real-time updates and notifications**
3. Facilitate **secure parent-teacher communication**
4. Improve **transparency in academics, attendance, and finance**
5. Deliver **mobile-first, user-friendly experience**
6. Integrate all major school services into one interface

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3. SYSTEM ROLES & ACCESS CONTROL

Role	Permissions
Parent/Guardian	View child data, communicate with school, make payments
Admin	Configure portal features and permissions
Teacher	Respond to parent messages, share updates
Finance Officer	Manage fee visibility and payments
Auditor	Read-only access to records

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4. MODULE ARCHITECTURE

```
Parent Portal
│
├── Dashboard
├── Academics View
├── Attendance Tracking
├── Finance & Payments
├── Communication Center
├── Events & Notifications
├── Health Records
├── Transport Tracking
├── Cafeteria Services
├── Settings & Profile
```

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5. DASHBOARD (REAL-WORLD DESIGN)

5.1 Key Widgets

- Student Overview (name, class, status)
- Attendance Summary (present, absent, late)
- Academic Snapshot (recent grades, performance trend)
- Fee Balance & Due Dates
- Upcoming Events (next 7 days)
- Notifications & Alerts

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6. ACADEMICS VIEW

6.1 Features

- Subject-level performance
- Exam results and report cards

- Teacher comments
- Performance trends over time

Upgrade from Current State

Instead of just:

“Average Grade: B”

You provide:

```
Math: A
English: B+
Science: A-
Trend: Improving
Teacher Comment: "Strong progress in problem solving"
```

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7. ATTENDANCE TRACKING

7.1 Features

- Daily attendance records
- Late arrivals and early departures
- Absence reasons
- Attendance trends

Real-World Rule

```
IF attendance < 85%
→ Notify parent immediately
→ Suggest meeting with school
```

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8. FINANCE & PAYMENTS

8.1 Features

- Fee structure breakdown
- Payment history
- Outstanding balances
- Due dates and reminders

8.2 Payment Integration

Supports:

- Mobile money (e.g., M-Pesa)
- Bank transfer
- Card payments

Flow

```
Invoice generated (Finance Module)
  ↓
Displayed in Parent Portal
  ↓
Parent makes payment
  ↓
Receipt generated
  ↓
Confirmation notification sent
```

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9. COMMUNICATION CENTER (CRITICAL UPGRADE)

9.1 Features

- One-to-one messaging (Parent ↔ Teacher)
- Group/class announcements
- School-wide notifications

Integration

Uses your **In-App Communication Module**

Real-World Flow

```
Teacher sends message
  ↓
Parent receives push notification
  ↓
Parent replies in-app
  ↓
Conversation logged for audit
```

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10. EVENTS & NOTIFICATIONS

10.1 Features

- Upcoming school events
- PTA meetings
- Exams and deadlines

Notifications

- Push notifications
- SMS/email fallback
- In-app alerts

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11. HEALTH RECORDS

11.1 Features

- Medical history
- Allergies
- Vaccination records
- Clinic visits

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12. TRANSPORT TRACKING

12.1 Features

- Bus route information
- Pickup/drop-off times
- **Live GPS tracking (recommended for real-world use)**

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13. CAFETERIA SERVICES

13.1 Features

- Daily/weekly menu
- Meal plans

- Payment integration

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14. MOBILE-FIRST DESIGN

14.1 Requirements

- Fully responsive UI
- Dedicated mobile app (Android/iOS)
- Offline caching (for poor connectivity areas)

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15. NOTIFICATIONS ENGINE

15.1 Triggers

- Fee due reminders
- Absentee alerts
- New grades posted
- Event reminders
- Emergency announcements

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16. SECURITY & COMPLIANCE

- Secure login (2FA recommended)
- Data encryption
- Role-based access (parent sees only their child)
- Compliance with data protection laws

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17. SCALABILITY & PERFORMANCE

- Support thousands of parents concurrently
- Multi-child support per parent
- Multi-campus compatibility
- Optimized API responses for mobile